



Differentiated instruction in Turkish classrooms: Practices, teacher experiences and barriers

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ABSTRACT

Despite the growing body of research on differentiated instruction (DI), the experiences of teachers outside of Anglo-European contexts remain underexplored. This study, based on the DI-Quest model, examines Turkish teachers' self-reported DI-Quest data, its links with demographic characteristics, and barriers to DI practices. Using a mixed methods sequential explanatory design, quantitative data from 472 teachers (470 valid forms) and interviews with 12 teachers in İzmir were analysed. Findings reveal that private school teachers implement DI more frequently, facing fewer barriers due to better support systems, including smaller classes, adequate infrastructure, professional development, diverse resources, supportive management, and parent collaboration. Interestingly, DI practices decline among more experienced teachers. Qualitative insights identify significant obstacles, such as intensive curricula, exam pressure, overcrowded classrooms, time constraints, inadequate resources, administrative challenges, and resistance from colleagues. The study underscores the multifaceted challenges within and beyond schools, emphasizing the need for systemic support. Contextual factors affecting DI in Türkiye are highlighted, urging targeted interventions.

1. Introduction

The global demand for inclusive, equitable, and high-quality education has gained increasing recognition, with growing advocacy for equal learning opportunities for all students (UN, 2015; UNESCO, 2017). Recent global challenges, including the COVID-19 pandemic, migration, and conflicts, have intensified student diversity in classrooms, underscoring the inevitability of unforeseen educational challenges. Differentiated Instruction (DI) has emerged as a promising pedagogical approach for addressing this diversity, enabling teachers to tailor instruction to students' varied learning needs (Letzel-Alt & Pozas, 2023). DI requires teachers to proactively modify their instruction in response to differences in students' prior knowledge, skills, interests, and learning profiles (Tomlinson, 2014). Although DI originated in the United States, it has gained traction worldwide, with studies documenting its implementation in Europe (Compen, Verstegen, Maussen, Hülsman, & Dolmans, 2024), Asia (Grecu, 2023), Africa (De Jager, 2017), Latin America (Pozas, Tovar, Guerra, Armendariz, & Zubiria, 2023a), and Australia (Gibbs & McKay, 2021).

Empirical research consistently highlights the positive impact of DI on student achievement and motivation, with meta-analyses reporting small to moderate effect sizes (Deunk, Smale-Jacobse, de Boer, Doolaard, & Bosker, 2018; Smale-Jacobse, Meijer, Helms-Lorenz, & Maulana, 2019). Similar findings have been reported in Türkiye (Bal, 2023; Taş & Minaz, 2024), where DI was adopted through educational borrowing. However, the complexities of international borrowing suggest that educational practices do not always transfer seamlessly between contexts (Heng, 2023; Phillips & Ochs, 2003). Educational ideas are often shaped by the context in which they arise, and the success of 'borrowed' practices depends on their adaptation to local cultural and educational environments (Grimsæth & Hallås, 2016).

One of the significant challenges associated with DI is its practical implementation. Notably, most research on DI has been conducted in Anglo-European contexts, with over half of the relevant literature originating from the United States, the United Kingdom, and Australia (Sun & Xiao, 2021). This geographic concentration highlights the need for studies exploring DI in non-Anglo-European settings to examine how teachers interpret, adapt, and implement DI in diverse educational

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landscapes. Moreover, while DI is often examined in terms of its benefits for students, its impact on teachers remains underexplored (Pozas, Letzel-Alt, & Schwab, 2023b). Research indicates that DI is not widely practiced in classrooms (Smit & Humpert, 2012) and that various barriers—including insufficient training (Schwab & Woltran, 2023), inadequate resources, large class sizes, and demanding curricula (Çulhaoğlu & Letzel-Alt, 2023; Davis, 2020; Heng, 2023)—limit its effective implementation. Despite these challenges, failing to offer DI can negatively affect teacher well-being, leading to lower job satisfaction and higher stress levels (Schwab & Woltran, 2023). Thus, in contexts beyond Anglo-European countries, studies are needed to look more closely at what is needed for teachers to be able to implement DI skillfully without perceiving it as an additional burden.

In Türkiye, as in other countries, teachers face significant obstacles in implementing DI (Aydoğan-Yenmez & Özpinar, 2017; Demirkaya, 2018). However, research on DI praxis in Türkiye remains limited (Çam & Acat, 2023; Kozikoğlu & Bekler, 2018), partly due to the absence of validated instruments to assess teachers' experiences with DI. The Differentiated Instruction Questionnaire (DI-Quest), theoretically developed by Coubergs, Struyven, Vanthournout, and Engels (2017) in Belgium and adapted into Chinese (Bi, Struyven, & Zhu, 2023; Yuen, Leung, & Wan, 2022), Hebrew (Dorfberger & Eyal, 2023) and Turkish (Tepetaş Cengiz, Tezcan, & Doğan, 2021), can be used to reveal the landscape of DI practices in diverse contexts. Findings derived from DI-Quest can provide valuable insights into DI implementation in Türkiye and inform strategies for addressing its associated challenges.

This study aims to examine the extent to which Turkish middle school teachers implement DI and to explore their perceptions of the barriers they encounter. By employing DI-Quest for the first time in Türkiye and adopting a mixed-methods approach, this study offers a comprehensive analysis of DI practices in a non-Anglo-European context. The findings contribute to the global discourse on DI implementation by identifying the specific challenges faced by Turkish teachers and offering practical recommendations for educators, administrators, and policymakers. Beyond providing empirical evidence, this research highlights broader implications for the successful integration of DI in culturally diverse settings, with key insights into professional development, policy formulation, and future research.

1.1. Research questions

Drawing on the DI-Quest model, this study aims to explore Turkish teachers' self-reported DI-Quest data, the links between these data and various demographic characteristics, as well as the barriers that hinder their DI practices. This leads to the following research questions:

1. How do Turkish middle school teachers implement DI, as reflected in DI-Quest data and their self-reported experiences?
2. Do teachers' DI-Quest scores vary by teaching experience, education level, subject area, class size, school type, and weekly teaching hours?
3. What key factors contribute to the challenges of DI implementation, as perceived by teachers?

1.2. Differentiated instruction (DI)

DI is a student-centered approach in which teachers flexibly modify elements in the learning environment to maximize learning opportunities for all students (Compen et al., 2024). For instance, diversity is offered in the level of complexity in instruction (content), the grouping of students (process) and the type of assessment (product) (Tomlinson, 2014). Teachers may implement DI through various instructional techniques, such as tiered assignments or flexible student groupings based on performance or interests (Pozas et al., 2023b). Rooted in social constructivist theory, DI aligns with the concept of the zone of proximal development (Vygotsky, 1978), emphasizing scaffolding and tailored

support to enhance student learning. Additionally, proactive instructional planning and formative assessment are essential elements of DI, enabling teachers to continuously adapt instruction based on student progress (Compen et al., 2024).

The DI-Quest framework (Coubergs et al., 2017) provides a structured model for assessing the extent to which teachers conceptualize and implement DI principles. It comprises five key dimensions: teachers' mindset, ethical compass, flexible grouping, output=input, and adaptive teaching. Coubergs et al. (2017) found that teachers with a growth mindset believe that skills, intelligence and ability are changeable and that this positively predicts DI. On the contrary, teachers with a fixed mindset believe that success is determined by ability and intelligence and tend to use controlled teaching in the classroom (Bi et al., 2023). The ethical compass refers to whether teachers see task completion or student development as the compass for teaching. In other words, teachers who emphasize faithful implementation of the curriculum to the neglect of student needs are less likely to implement DI (Yuen et al., 2022). Flexible grouping refers to grouping students homogeneously and heterogeneously according to objectives and changing this flexibly so that students experience both independent learning and working in a variety of cooperative groups (Coubergs et al., 2017). The term output=input underscores the role of continuous assessment in guiding instructional adjustments (Bi et al., 2023), while adaptive teaching involves modifying instruction based on students' readiness, interests, and learning profiles to enhance engagement and motivation (Tomlinson, 2014).

1.3. DI in the Turkish context

For several years, Türkiye has been challenged by the demand for more inclusive classrooms. Hence, DI research over the past decade has primarily focused on student achievement (Karadağ, 2014; Taş & Minaz, 2024), with limited attention to teachers' experiences in implementing DI. Major obstacles identified include time constraints, resource limitations, extensive curricula, large class sizes, insufficient teacher training, and a lack of institutional support (Piltan, 2016). Although studies have addressed the challenges faced by various teacher groups—such as mathematics (Aydoğan-Yenmez & Özpinar, 2017), classroom (Çulhaoğlu & Letzel-Alt, 2023; Gülay, 2021), and preschool teachers (Özkanoglu, 2015)—the depth of teachers' practical experiences with DI remains underexplored.

Recent developments indicate a rising interest in DI in Türkiye. Both pre-service and in-service teacher training programs increasingly emphasize DI, and the 2024 revisions to the Turkish national curriculum include explicit guidance on its implementation (MoNE, 2024). Given these advancements, DI is expected to gain more prominence in educational policy and practice, making this study particularly timely in informing future developments.

2. Method

To elucidate the research problem both broadly and in depth, the sequential explanatory design (Creswell & Plano-Clark, 2018) of mixed methods was used in this study. Firstly, based on the DI-Quest model, Turkish middle school teachers' teaching experiences, level of education, subject area, class size, type of school and number of weekly teaching hours were identified in relation to their self-reported DI practice, and secondly, interviews were conducted to support and understand the quantitative data in more depth.

2.1. Participants

Considering the sample size to represent the population by random sampling, data should be collected from at least 379 teachers (Gay & Airasian, 1996) out of 6506 teachers (MoNE, 2022) working in six subjects (Turkish, Mathematics, Science, Social Studies, English,

Religious Culture and Ethics-RCE) in the central districts of Izmir, the third most populous city in Türkiye, in the 2022–2023 academic year. The quantitative data were collected from 472 teachers and analyzed on 470 forms (335 females, 134 males, 1 other). (For the demographic characteristics of the first sample, please see Appendix 1). In the second phase of the study, after the analysis of the quantitative data, the second study group was selected to be interviewed to deepen the data. Using the maximum diversity sampling, interviews were conducted with 12 volunteer teachers (see Appendix 2). The scores of the teachers who participated in the interviews from DI-Quest instrument are given in Appendix 3.

2.2. The DI-Quest instrument

The DI-Quest Instrument consists of five constructs (see Table 1): growth mindset, ethical compass, flexible grouping, output=input, and adaptive teaching (Coubergs et al., 2017). The first two constructs represent teachers' philosophies of DI, while the next two pertain to principles for organizing DI teaching. The final construct, adaptive teaching, involves DI based on students' interests, readiness, and learning profiles (Gheysens, Couberts, Griful-Freixenet, Engels, & Struyven, 2022). Couberts et al. (2017) conducted exploratory and confirmatory factor analyses (EFA and CFA) to assess the DI-Quest's factor structure, resulting in a five-factor model with 31 items demonstrating a good fit (CFI=0.919;TLI=0.911;RMSEA=0.041 [0.037–0.044, 90 % CI, $p(0.05) = 1.000$]; SRMR = 0.048; $\chi^2 = 5888.338$, $df = 465$, $p = 0.000$). Cronbach's alpha values for the factors range from 0.63 to 0.86.

The DI-Quest instrument was adapted into Turkish by Tepetaş Cengiz et al. (2021), following stages of translation, semantic clarification, expert review, back-translation, piloting, and finalization. After adapting the scale, the 31-item form was administered online to 603 volunteer teachers. Two rounds of EFA reduced the instrument to 25 items while maintaining the original five-factor structure. The revised constructs included three items for growth mindset, five for ethical compass, eight for flexible grouping, three for output=input, and six for adaptive teaching. Cronbach's alpha coefficients were 0.80, 0.75, 0.73, 0.83, and 0.86, respectively. CFA confirmed the five-factor structure with good fit values ($\chi^2/df=1.60 < 2.0$, GFI=0.90, SRMR=0.051 < 0.10, RMSEA=0.045 < 0.05, CFI=0.96 > 0.900, NNFI=0.96 < 1.00). The results of the CFA on the data collected in this study were in line with the findings of Tepetaş Cengiz et al. (2021) ($\chi^2/df=2.026 < 3$, RMSEA=0.047 < 0.05, SRMR=0.05 = 0.05, CFI=0.937 > 0.900, TLI=0.930 > 0.900). The internal consistency coefficient of growth mindset was 0.89; ethical compass was 0.77; flexible grouping was 0.84; output=input was 0.83; adaptive teaching was 0.89.

In cross-linguistic/cross-cultural scale adaptation studies, some items may work differently from the original scale due to cultural

variations. Tepetaş Cengiz et al. (2021) also discarded some items because they loaded on more than one construct. The factor structure remained unchanged, even though the number of items in the constructs had been altered. Although negative results may occur in cases where the removal of some items affects content validity, Tepetaş Cengiz et al. (2021) did not mention such a situation. Undoubtedly, the extent to which omitted items affect results in terms of construct and content validity could be the focus of further research. However, as mentioned above, the reliability coefficients of the constructs in this study range between 0.77 and 0.89, which seems sufficient.

2.3. Interviews

After analyzing the quantitative data, a previous study (Aydoğan-Yenmez & Özpınar, 2017) was used for the questions in the semi-structured interviews, and then a form was drafted after receiving criticism and suggestions from two experts. The clarity and order of the questions were checked, and some minor changes were made after two pilot interviews of about 30 min each. Following the warm-up questions, the interviews included questions about how much and how DI was used, whether any training related to DI was attended, which DI techniques were most used, how techniques were chosen, difficulties encountered in implementing DI, working with other colleagues, and what it takes to implement DI effectively.

2.4. Data collection

After obtaining Research Ethics Committee of Ege University (approval number and date: 1641/2022) and MoNE approval, quantitative data were collected from 470 teachers during the autumn semester of the 2022–2023 academic year through both face-to-face and online methods. Teachers were reached via school administrations, with the support of principals who helped distribute the questionnaire links or paper forms during departmental meetings or through internal communication channels. Participants signed a printed/online informed consent form, which was accompanied by an information letter outlining the procedures for data collection, analysis, storage, and reporting. At the end of the DI-Quest, volunteers were informed about subsequent interviews and invited to provide their contact details. In the spring semester, qualitative data were gathered through face-to-face interviews with 12 volunteers who completed the DI-Quest. To ensure comfort, the interview settings and times were arranged with the participants, and the sessions were audio recorded. Each interview lasted approximately 35/40 min, totaling 420 min and seven seconds.

2.5. Data analysis

A multivariate analysis of variance (MANOVA) was initially planned to analyze the DI-Quest scores, but skewness and kurtosis (0.854–3.932) indicated non-normality. Therefore, non-parametric tests—Mann-Whitney *U* and Kruskal-Wallis *H*—were used along with descriptive statistics. Mann-Whitney *U* compared groups with two categories (e.g., school type, education level), while Kruskal-Wallis *H* was applied for groups with more than two categories (e.g., teaching experience, subject area). Effect sizes were interpreted using Pearson's *r* (0.1–0.3 small, 0.3–0.5 medium, >0.5 large) and η^2 (0.01–0.06 small, 0.06–0.14 medium, >0.14 large) (Cohen, 1988).

The thematic analysis suggested by Braun and Clarke (2006) was used to analyze the qualitative data. Firstly, the audio recordings were transcribed and filed for each teacher (116 pages in total). After the files were uploaded to MAXQDA 2020, the first author drafted a code list after thorough reading, which was refined in collaboration with the second author by merging, renaming, and removing unnecessary codes. The coded data were divided into categories and themes, and their relationships to each other, whether consistent or not, were examined and named. At this stage, an effort was made to maintain internal

Table 1
The description of DI-Quest instrument (Couberts et al., 2017).

Constructs	Dimensions	Number of items	Example items
Philosophies of DI	Growth mindset	5	Despite the learning potential of students, we cannot change their intellectual capacities
	Ethical compass	6	The curriculum does not provide any room to foresee in differentiated instruction
Principles of DI	Flexible grouping	8	I explicitly make sure I have a good relationship with all my students
	Output=input	4	I teach my students how to cope with feedback
The practice of DI	Adaptive teaching	8	During my lessons, different students work on different tasks with different levels of difficulty

homogeneity by not transferring codes and categories from one topic to another, as well as external heterogeneity (Miles & Huberman, 1994) by ensuring that the themes developed were distinct, clear, unambiguous, and intelligible. To achieve this, theme names were iteratively revised for clarity and checked for overlap or conceptual redundancy. When two candidate themes were similar, the researchers revisited the coded excerpts to test whether they conveyed sufficiently different meanings to justify separate thematic labeling. Finally, after the codes and themes were reviewed, direct quotations were determined. Codes, categories and themes are presented in Appendix 4.

2.6. Validity, trustworthiness and reflexivity

To address validity threats in the explanatory sequence design identified by Creswell and Plano-Clark (2018), qualitative data were collected through maximum diversity sampling aligned with DI-Quest findings. In the first stage, participants were informed about the second stage of data collection, and consent was obtained. Additionally, we employed various strategies based on Lincoln and Guba's (1985) framework to enhance credibility and trustworthiness. Firstly, the transcriptions were member-checked to ensure credibility, and expert opinions and pilot interviews were used for the interview questions. The codes and categories obtained were compared at different time intervals to ensure consistency. To support transferability, the research design, participants, instruments and data analyses were explained in detail; raw data and coding were stored. The second author, who had more research experience, reviewed the first author's initial analyses for peer debriefing and checked the findings from the interviews for accuracy. Two external researchers also provided peer debriefing for the researchers' analyses. Finally, we discussed all steps of the data analysis together to ensure that all decisions were validated.

The first researcher has insights into the DI practices enacted in schools as she is a science teacher with eight years of experience working in a public middle school. However, when interviewing, she was careful not to add her own interpretations about DI practices, not to guide, and to act impartially, and she encouraged the participants to feel comfortable expressing themselves by being friendly and sincere. Moreover, the second author helped to identify blind spots in the study, while two other external reviewers contributed to the peer review process, which facilitated cross-checking of the findings and validation of the results.

3. Findings

3.1. Turkish middle school teachers' DI practices

The findings related to the first question of the study are presented through the integration of quantitative and qualitative data. Firstly, descriptive statistics of the scores obtained from the DI-Quest Instrument (Table 2), and then the joint display of quantitative and qualitative findings are presented (Fig. 1).

Table 2 shows that the highest scores in DI-Quest are in output=input and growth mindset. This indicate that Turkish middle school teachers have a growth mindset in DI practices and plan instruction in accordance with student performance. It is also noteworthy that the lowest scores were obtained from ethical compass. Fig. 1 presents quotes from teachers with high and low scores across the five DI-Quest

constructs, clarifying the practical implications (shaded texts indicate low scorers). The quantitative and qualitative findings support and complement each other. For instance, T8, who scored high in adaptive teaching, describes how they account for students' diverse characteristics in their instruction. Similarly, T5, with a high output=input score, explains how he utilizes test results to support student learning and development, confirming the high mean scores in this dimension. Corresponding quotes are also available for teachers with low scores.

3.2. The links between teachers' DI-Quest instrument scores and demographics

The Mann-Whitney U results indicated no statistically significant differences in teachers' growth mindset and ethical compass scores based on education level, school type, or weekly instructional hours. While flexible grouping scores did not differ significantly by education level or instructional hours, there was a notable difference based on school type (Table 3). Specifically, teachers in private schools scored higher in flexible grouping than their public-school counterparts, although the difference was small ($r = 0.205 < 0.3$). Significant differences were observed in output=input scores based on teachers' educational level, school type, and weekly course load; however, the effect size was small. While adaptive teaching scores did not significantly differ by education level or weekly load, they did vary by school type, with private school teachers scoring higher than public school teachers ($U=14892.0$; $p < 0.001$), though the effect was small ($r = 0.178 < 0.3$).

Kruskal-Wallis H results indicate that growth mindset scores did not differ statistically by teaching experience, branch, or class size. However, ethical compass scores showed a significant difference by branch ($H=30.440$; $df=5$; $p < 0.001$; $\eta^2=0.055 < 0.06$), while remaining consistent across years and class sizes (see Table 4). The Bonferroni test revealed that Turkish language teachers scored significantly lower than social studies ($TS=74.283$; $p = 0.016$), science ($TS=83.908$; $p < 0.001$), and RCE teachers ($TS=113.205$; $p < 0.001$). Although the scores for flexible grouping were not statistically different according to seniority and branch, there was a statistically significant difference between at least two groups according to class size. Bonferroni tests reveal that teachers in classes with 20 or less students were more flexible in grouping students according to their individual needs compared to overcrowded classes ($TS=58.895$; $p = 0.002$).

In terms of output=input scores, although there is no statistical difference in seniority and branch, there is a significant difference between at least two groups according to class size, and the effect of this difference is small. Bonferroni tests show that the output=input scores of the group with 20 or less students were significantly higher than the scores of the group with 21–30 students ($TS=39.171$; $p = 0.030$). When class sizes are small, teachers do more formative assessments and monitoring, considering students' individual needs.

There are statistically significant differences in teaching experience, branch and class size for adaptive teaching, but the effect size is small. Bonferroni tests indicate that differences favor novice teachers (0–5 years) and English language teachers, who differentiate instruction according to students' interests, readiness, and learning profiles more frequently than their more experienced and mathematics counterparts. Additionally, teachers in crowded classes demonstrated weaker DI practices compared to those with 20 or fewer students ($TS=58.898$; $p = 0.001$).

3.3. Key drivers of DI implementation challenges

The qualitative findings showed that Turkish middle school teachers face challenges in implementing DI due to a) the intensive curriculum contents and lack of curricular resources, b) performance pressure, c) school context (excessive course load, lack of administrative support, inadequate physical environment, overcrowded classrooms, extracurricular duties), d) teacher-specific reasons (lack of time for planning,

Table 2

Teachers' DI-Quest instrument scores by dimensions (n = 470).

DI-Quest Constructs	Median	Mean	SD	Min	Max
Growth mindset	6.00	5.74	1.273	1	7
Ethical compass	2.80	2.91	1.204	1	7
Flexible grouping	5.56	5.43	0.862	1	7
Output=Input	6.00	5.84	0.887	1	7
Adaptive teaching	5.67	5.41	1.007	1	7

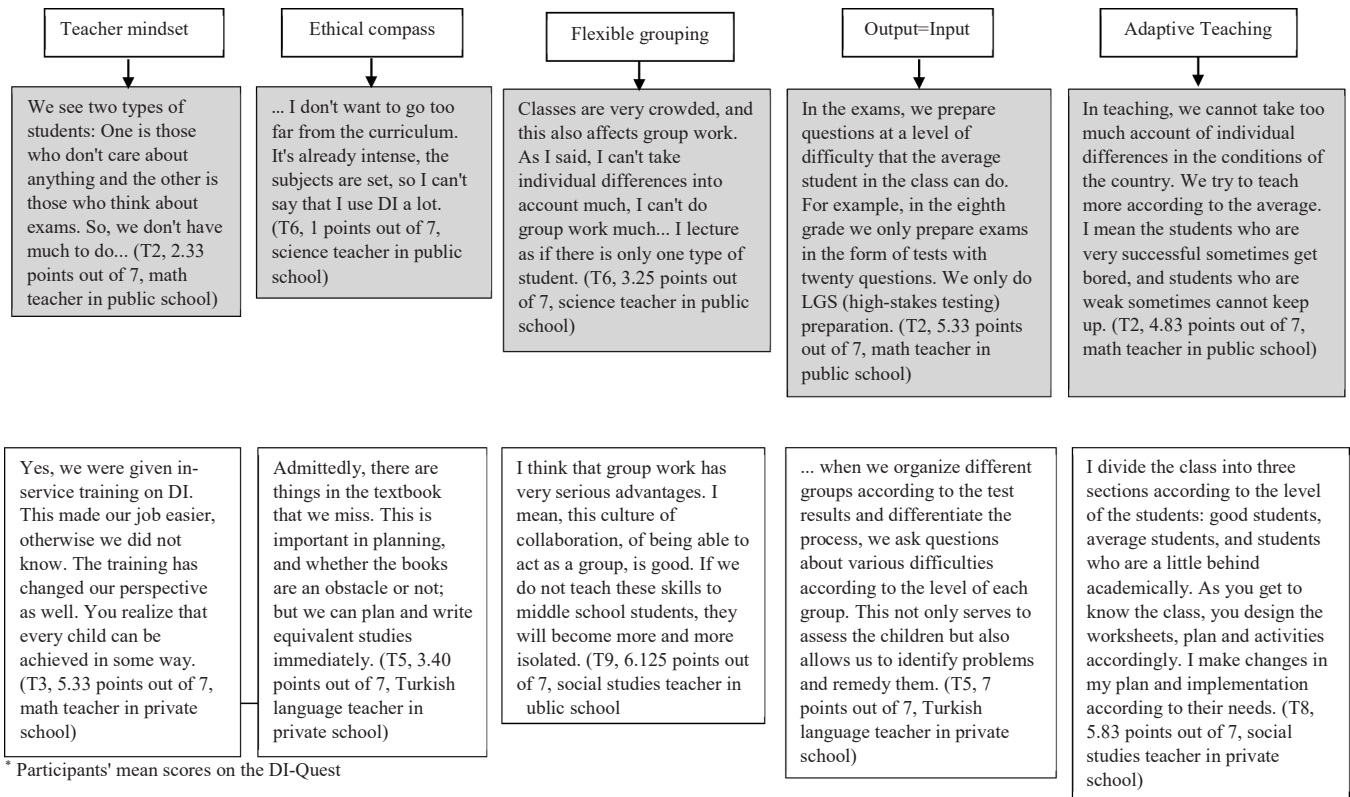


Fig. 1. Quotes of teachers with low and high DI-Quest mean scores (shaded texts indicate low scorers, others indicate high scorers*).

Table 3
Mann Whitney *U* findings on DI-Quest scores constructs.

Construct	Group variable	Categories	N	Median	Rank Mean	<i>U</i>	<i>Z</i>	<i>p</i>	<i>r</i>
Flexible grouping	School type	Public	361	44	220.24	14166.5	4.440***	< 0.001	0.205
		Private	109	46	286.03				
Output=Input	Education level	Bachelor	387	18	229.52	13748.0	2.118*	0.034	0.098
		Graduate	83	18	263.36				
	School type	Public	361	18	224.94	15863.5	3.153**	0.002	0.145
		Private	109	18	270.46				
Adaptive teaching	Weekly course load	21 or less	182	18	219.37	23273.0	-1.985*	0.047	-0.092
		22 +	286	18	244.13				
	School type	Public	361	33	222.25	14892.0	3.861***	< 0.001	0.178
		Private	109	35	279.38				

p* < 0.05; *p* < 0.01; ****p* < 0.001

classroom management difficulties, amotivation, resistance and obstructive attitudes of colleagues, lack of knowledge about DI), e) student reluctance and misbehavior, and f) problems with parents.

The reasons for teachers' difficulties with the curriculum and curricular resources are the lack of materials and textbooks, and the lack of encouragement for DI in textbooks published by the MoNE. Regardless of school type and branch, teachers reported that textbooks are far from providing guidance for DI practices. Combine this with the intensive curriculum content and lack of teacher guidance, implementation of DI in the classroom is entirely at the discretion of the teacher:

Frankly, we cannot do this (DI) very often. We have a curriculum to catch up on. We have a lot of (math) questions to answer. So, we can't say that we use [DI] that much. (T2, mathematics teacher working in a public school)

Teachers stated that they were under various pressures to improve course achievement and that these pressures prevented them from using DI. The exam-oriented education system, and in particular high-stakes testing (the national central exam at the end of middle school for

entry to qualified high schools), appears to be behind these pressures:

When the aim is success in the exam, education through lecture is at the forefront. Now, when the system is geared towards this, there is such an expectation from the parents, they want success in the exams. There is also such an expectation from the school administration. So, the question is always: has success increased? Have the scores gone up? (T1, mathematics teacher in a public school)

The school context problems are related to excessive course load, lack of administrative support, inadequate physical environment, overcrowded classrooms, extracurricular duties. Teachers in public schools reported that the school administration did not tolerate and support the noise that occurred during the implementation of DI. In private schools, on the contrary, support was strong:

...DI is our philosophy, so we are encouraged to do it in school. If you don't, there are problems, our lessons are monitored. Sometimes with notice and sometimes without, and they often give feedback on the teaching methods and techniques used. (T5, Turkish language teacher working in a private school)

Table 4Kruskal-Wallis H findings on DI-Quest scores constructs.

Construct	Group variable	Categories	N	Median	Rank Mean	H	df	p	η^2
Ethical compass	Branch	Turkish language	99	11.0	186.37	30.440***	5	< 0.001	0.055
		Mathematics	116	13.0	236.95				
		Science	85	15.0	270.28				
		Social Studies	56	15.0	260.65				
		English language	80	13.0	212.41				
Flexible grouping	Class size	RCE	34	17.5	299.57	12.144**	2	0.002	0.022
		20 or less	111	45.0	263.00				
		21–30	211	44.0	233.11				
		31 +	140	43.0	204.10				
Output=Input	Class size	20 or less	111	18.0	258.95	6.826*	2	0.033	0.011
		21–30	211	18.0	219.78				
		31 +	140	18.0	227.40				
Adaptive teaching	Seniority	0–5 years	32	35.0	299.97	15.063*	6	0.020	0.021
		6–10 years	88	33.5	233.56				
		11–15 years	79	33.0	207.94				
		16–20 years	116	34.5	252.90				
		21–25 years	72	32.5	211.20				
		26–30 years	52	33.0	229.63				
		31 +	31	34.0	245.84				
	Branch	Turkish language	99	34.0	238.22	16.822**	5	0.005	0.026
		Mathematics	116	33.0	202.07				
		Science	85	33.0	218.76				
		Social Studies	56	34.5	262.24				
		English language	80	35.0	262.41				
		RCE	34	35.0	276.13				
	Class size	20 or less	111	34.0	264.27	12.129**	2	0.002	0.022
		21–30	211	34.0	231.60				
		31 +	140	32.5	205.37				

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Similarly, teachers working in public school's state that classrooms are small, and this makes it difficult to implement DI, while those working in private schools do not experience such a problem. The same is also true regarding class sizes (see quotes below). These qualitative findings confirm the statistically significant differences found between school types and class sizes in the dimensions of flexible grouping, output=input, adaptive teaching.

There are classes where sixty or forty or forty-five people are trying to learn together. The number of students is important, of course. Not only if you use a different method, but also if you want to make an ordinary lecture, the excessive class size prevents that. (T1, mathematics teacher working in a public school)

It was found that both the weekly course load and teachers' extra-curricular duties made it difficult for them to implement DI. In particular, the complaints of teachers working in private schools on this issue seem to be consistent with the quantitative findings (in terms of output=input scores) regarding school type and weekly course load.

...we have a lot of paperwork. There is also this: I am responsible for Technofest, I am responsible for Tübitak (science project), I have a robotics coding task, so you are scattered. Of course, I cannot mentally concentrate on such methods because I have too many things to train. (T7, science teacher in a private school)

Teachers' lack of time to plan DI, classroom management challenges, colleagues' resistance and obstructive attitudes towards DI, amotivation and lack of knowledge about DI are among the notable qualitative findings. Disruptive pupil behavior, classroom management problems and the obstructive attitudes of colleagues reduce motivation. Indeed, the quantitative findings in relation to adaptive teaching and teacher seniority showed that novice teachers implemented DI more, and the amotivation that occurred over time was also highlighted:

I think DI is not implemented in most public schools. Maybe the new teachers (implement it), which I tried a lot when I first started working. Younger teachers are more enthusiastic. But over the years, because we work in poor conditions in public schools and our efforts

are not appreciated, teacher burnout happens... (T6, science teacher working in a public school)

Teachers' lack of knowledge and training in DI is also an important problem. A mathematics teacher (T2) explained this as follows: *Anyway, these approaches were not explained to us in pre-service education. Maybe we do not use them because we do not know them completely.* There were also teachers (T7) who underlined the pupils' reluctance or resistance to participate in the lesson: *Actually, the children seem to be very eager to do something different, but they are also influenced by their families and say, "can we solve a test instead of doing this or can I solve a test"?* Finally, some obstructive attitudes of parents are also noteworthy. Accordingly, it was observed that some parents' pressure to succeed in high-stakes testing and their expectations of traditional teaching made it difficult to implement DI. Those working in private schools (T8, social studies teacher) did not mention this issue: *The management of our school tries to involve parents. As the school's approach has been gamification since primary school, practices such as DI seem very natural to parents.*

4. Discussion

The quantitative findings reveal that the lowest factor mean for ethical compass aligns with studies from Belgium (Coubergs et al., 2017), China (Bi et al., 2023), Hong Kong (Yuen et al., 2022), and Israel (Dorfberger & Eyal, 2023). The low ethical compass scores may reflect teachers' reluctance to deviate from the curriculum, focusing primarily on covering topics. Recent studies also indicate high fidelity to the curriculum among Turkish teachers (Aslan & Erden, 2020; Karakuyu & Oğuz, 2021). In contrast, the highest factor mean for output=input is comparable only to findings from China and Hong Kong. Although the Turkish version of the DI-Quest instrument was reduced to 25 items through factor analyses during adaptation (Tepetaş Cengiz et al., 2021), the five original constructs were retained. As such, construct-level comparisons with international studies remain largely valid. However, due caution is advised when comparing total scores across countries, as differences in item count and structure may influence score distributions.

Teachers in public schools faced more challenges in implementing DI compared to those in private schools. Scores for most DI-Quest constructs were higher in private schools. Overcrowding in Turkish public schools (Eramil & Kasalak, 2024) likely explains the significant advantage of private schools in flexible grouping. Public school teachers reported that inadequate physical infrastructure and seating arrangements hindered group work, issues not mentioned by private school teachers. Previous Turkish studies (Çulhaoğlu & Letzel-Alt, 2023; Demirkaya, 2018) indicate that private school teachers excel in DI practices. Thus, private schools appear more aligned with DI philosophy and better at addressing related challenges. The corridors and classrooms of private schools in Türkiye are different from those of public schools—not only physically, but also in terms of the pedagogical mindset—mainly because they closely follow Anglo-European pedagogy. However, the increase in private schooling rates (5.74 % in primary, 6.51 % in middle, and 10.36 % in high school level) in recent years (MoNE, 2022) is quite alarming in terms of inclusive education and equity. This trend underscores why DI must not be limited to well-resourced private schools but prioritized as a core strategy in public education. Given that public schools serve most students, including those from disadvantaged backgrounds, ensuring effective DI in these settings is essential for promoting both educational equity and quality.

While the current findings highlight a clear contrast between public and private schools, qualitative data also suggest that meaningful differences may exist within each school type. For instance, some public school teachers described extreme overcrowding and lack of administrative flexibility, while others mentioned slightly more manageable conditions. Similarly, although private schools generally offer more favorable settings for DI, several teachers reported additional responsibilities—such as coordinating extracurricular projects or competitions—that limit their instructional planning time. These nuances indicate that DI implementation is influenced not only by school type but also by internal factors such as school leadership, organizational culture, and workload distribution.

The discovery that teaching experience and class size positively predict DI practices (Bi et al., 2023) aligns with this study's results, which show flexible grouping, output=input, and adaptive teaching factors significantly favor smaller class sizes. Yet, it contradicts the finding that adaptive teaching favors teachers with 0–5 years of experience. Previous research has also shown mixed results regarding the relationship between teaching experience and DI practices. Some studies indicate that teachers with over five years (Bi et al., 2023) or eight years (Sheehan, 2011) of experience practice DI more frequently, while others found no significant correlation (Donnell & Gettinger, 2015; Merawi, 2018). Our qualitative findings suggest that although more experienced teachers might be expected to apply DI more successfully, a lack of support within the school setting can diminish motivation after negative experiences. These discrepancies may also reflect cross-national differences in teacher preparation programs, policy expectations, and monitoring mechanisms. For example, countries with more practice-oriented training, policy-level encouragement for inclusive pedagogies, and supportive forms of instructional supervision may create more favorable conditions for sustained DI use across a teacher's career.

Wen and Cai (2024) identified flexible grouping as a positive predictor of DI practices, while Bi et al. (2023) noted the difficulty teachers face in implementing flexible grouping in large classes. This study found significant differences in the flexible grouping scores, favoring smaller classes (20 or fewer students). Given the similar statistical outcomes for output=input and adaptive teaching, supported by qualitative findings, overcrowded classrooms emerge as a major obstacle to DI practices. Overcrowded classrooms have a negative impact on the implementation of DI (Gülay, 2021; Demirkaya, 2018; Heng, 2023). Moreover, the challenges of flexible grouping due to inflexible furniture and seating arrangements in Singaporean classrooms (Heng, 2023) also seem applicable to Turkish teachers.

The fact that there were no significant differences in the results of DI-

Quest scores by branch in general confirms the findings of Çam and Acat (2023) on Turkish middle school teachers. However, in this study, it was found that English teachers differentiated instruction more frequently than mathematics teachers. This may be since mathematics teachers use more traditional and/or test-oriented teaching methods. Indeed, recent studies in Türkiye focus on realistic mathematics teaching and highlight issues related to the tendency of test-oriented teaching due to centralized exams (e.g. Duran, Berk, & Köklü, 2023). Our qualitative findings on performance pressure also seem to be related to the test/exam-oriented nature of teaching due to high-stakes testing. These findings are aligned with studies (Demirkaya, 2018; Heng, 2023; Yuen et al., 2022) that exam-oriented educational contexts make DI more difficult. This discrepancy may be further explained by the ways in which different subjects are positioned within the national education system. In many contexts, mathematics is treated as a high-stakes, 'gatekeeping' subject, tightly linked to standardized assessments and future academic mobility (Battey, 2013; Stinson, 2004). Consequently, mathematics teachers may feel constrained by rigid curricular pacing, narrow performance metrics, and external accountability demands, leaving limited room for pedagogical flexibility. In contrast, English teachers—especially those exposed to international pedagogical models—may enjoy relatively more curricular space and professional autonomy, which facilitates the use of DI strategies. This highlights how subject culture, curricular expectations, and assessment pressures interact to shape teachers' instructional decisions.

The importance of curricular resources in facilitating DI practices is well recognized (Compen et al., 2024; Çulhaoğlu & Letzel-Alt, 2023; Schwab & Woltran, 2023) and supported by the qualitative findings of this study. Similarly, the barriers created by excessive content in curricula have been highlighted in previous study in Türkiye (Çulhaoğlu & Letzel-Alt, 2023). A study in Singapore also found that teachers felt constrained by centralized curricula and standardized national examinations, with little room to adjust content and product (Heng, 2023). Thus, the management and scope of curricula and teacher autonomy have once again re-emerged as determinants of DI practices.

Previous studies identify classroom management as a key challenge in implementing DI (Compen et al., 2024; Özkanoglu, 2015; Çulhaoğlu & Letzel-Alt, 2023; Gülay, 2021), a finding also supported by this study's qualitative data. Similarly, student reluctance (Çulhaoğlu & Letzel-Alt, 2023) and disengagement (Demirkaya, 2018) align with the experiences of public-school teachers in this study. Additionally, teacher amotivation, linked to the declining status of the profession and burnout, negatively affects DI, especially among experienced teachers. Comparable findings in Türkiye (Demirkaya, 2018) and Germany highlight that DI challenges, such as preparation, student diversity, and lack of support, increase teacher workload, stress, and time pressure, ultimately reducing job satisfaction (Pozas et al., 2023b). These findings resonate strongly with the broader literature on teacher attrition and resilience, which emphasizes how unsupportive working conditions, limited autonomy, and high-stakes accountability contribute to emotional exhaustion and professional disengagement (Pozas et al., 2023b; Schwab & Woltran, 2023). The tendency of more experienced teachers to avoid DI may reflect cumulative effects of these factors over time. Conversely, promoting teacher resilience through ongoing support, recognition, and collaborative environments could foster greater engagement with DI, even under challenging circumstances.

Our qualitative findings align with studies showing that a lack of teacher collaboration hinders DI implementation (Davis, 2020; Demirkaya, 2018; Grecu, 2023; Smit & Humpert, 2012). Strengthening collaboration skills among Turkish teachers, who tend to have weaker teamwork skills (Özdoğru, 2021), is essential. While this appears to be a systemic challenge, some schools in Türkiye have been identified as 'positive outliers' that foster strong collaborative cultures through distributed leadership and shared instructional goals (Çoban, Özdemir, & Bellibaş, 2023). These schools demonstrate that when conditions are supportive, teacher collaboration can thrive even in the broader context

of limited systemic cooperation. Effective school principals promote a climate of trust and prioritize instructional collaboration, improving both teaching quality and student outcomes. Additionally, reducing teachers' extracurricular duties and keeping weekly course loads manageable is crucial for DI success (Heng, 2023). Finally, fostering better parent-teacher communication in public schools, modelled after successful practices in private schools, could address DI challenges involving parents (Çulhaoğlu & Letzel-Alt, 2023; Demirkaya, 2018; Gülay, 2021; Özkanoglu, 2015).

5. Conclusions

As the first study to explore DI practices among middle school teachers in Türkiye using a mixed-methods sequential explanatory design, this research explores an underexamined aspect of DI practices in Türkiye. Given the limited focus on DI borrowing outside Anglo-European countries (Bi et al., 2023; Heng, 2023; Wen & Cai, 2024) and the neglect of teachers' challenges with DI (Pozas et al., 2023b), this replication study in the Turkish context was necessary. By capturing the experiences of Turkish middle school teachers, this study contributes to the broader DI literature.

The DI-Quest results and interview data are consistent and complementary, offering a comprehensive view of Turkish middle school teachers' DI practices. The quantitative findings reveal the extent and nature of DI implementation, while the qualitative data shed light on the barriers teachers face in applying DI. Adaptive teaching—a core construct of the DI-Quest model and a direct indicator of DI—was significantly higher among teachers working in private schools. These teachers reported fewer barriers and more favorable conditions, including smaller class sizes, adequate infrastructure, greater access to professional development, diverse curricular resources, supportive management, and positive communication with parents. It is also interesting to note that teachers who are generally new to the profession (those with 0–5 years of teaching experience) are more likely to use adaptive teaching, while DI practices decrease as experience increases. Although the data indicate that both school type and teaching experience affect adaptive teaching, interaction effects between these two variables were not explored in depth. Nonetheless, qualitative findings suggest that the supportive climate in private schools may help sustain DI practices even among more experienced teachers. In terms of general qualitative findings, intensive curriculum content, inadequate resources, tools, equipment and physical infrastructure, pressure to succeed in exams, excessive course load, overcrowded classes, lack of administrative support, extracurricular tasks, lack of time, difficulties in classroom management, amotivation, resistance and obstructive attitudes of colleagues, lack of information about DI, student reluctance and lack of motivation, problems with parents are the challenges that make DI practice difficult.

6. Implications for policy and practice

This study offers several implications for educational policy and classroom practice in Türkiye. Recognizing that diversity is now the norm in classrooms, DI should be regarded not as an optional method, but as a core pedagogical principle of inclusive education (Hattie, 2023). Yet, effective DI depends not only on its formal presence in policy but on how it is implemented by teachers (Schwab & Woltran, 2023). The finding that DI practices are more frequently observed in private schools underscores enduring disparities in access to the structural and professional conditions needed by DI.

While the inclusion of DI in the 2024 national curriculum (MoNE, 2024) is a welcome development, translating this vision into practice requires targeted and sustained support. First, interactive, long-term professional development programs—including mentoring and follow-up mechanisms—should be swiftly launched. Curricula should be simplified, and schools should be provided with DI-related materials,

ready-made plans, and adaptable instructional resources. This is particularly crucial given that many barriers to DI implementation lie beyond teachers' control, including classroom overcrowding, rigid instructional timeframes, and lack of resources (Schwab & Woltran, 2023).

The observed decline in DI use among more experienced teachers further reinforces the need for continuous professional learning, as well as systemic recognition of teacher workload and burnout. School leaders—especially in public schools—play a key role in creating collaborative cultures that encourage instructional innovation and reduce resistance to DI. Finally, in Türkiye's exam-driven system, it is essential to raise parental awareness about the developmental benefits of DI and to co-construct expectations that support student growth, not just test performance.

By offering a context-sensitive, mixed-methods analysis from a country that has adapted DI from Anglo-European systems, this study contributes valuable insight into how national policy, school conditions, and teacher realities interact in shaping the practice of differentiation. These findings may inform both local reforms and broader international discussions on how to support equitable, responsive instruction in diverse learning environments. As many countries face increasing student diversity, standardized testing pressures, and policy borrowing from dominant education systems, this study illustrates how DI can be recontextualized and supported through systemic, sustained, and locally grounded strategies.

7. Limitations and further studies

Like all research, this study has some limitations. First, as the DI-Quest relies on self-reported data, some teachers may have provided socially desirable responses. Although individual interviews were conducted to deepen the findings, future studies may consider incorporating lesson observations or video recordings to mitigate this limitation. Additionally, as Letzel-Alt and Pozas (2023) note, there is a growing global need for more empirical studies on DI. Replication studies using the DI-Quest across different regions and educational contexts are therefore encouraged.

Since this study used a cross-sectional design, limiting causal interpretations, we recommend longitudinal and experimental studies to further explore DI practices in Türkiye. For instance, a longitudinal design could track novice teachers' use of adaptive teaching over the first five years of their careers, helping to identify when and why DI practices increase or decline. Experimental studies could investigate the impact of targeted professional development interventions—such as collaborative lesson planning, coaching, or AI-supported micro-teaching—on teachers' ability to implement DI effectively. These designs would offer deeper insights into the developmental processes and contextual factors influencing DI adoption. Moreover, as data were collected only from middle schools in Izmir, expanding the sample to include different grade levels and regions could provide a broader analysis. This study is also limited to middle school teachers in six core subjects, leaving room for future research on DI practices in areas like music, physical education, visual arts, and technology & design. Another fruitful direction would be to offer DI training programs and compare teachers' DI-Quest responses before and after the intervention. To maximize impact, such programs could be conducted at the school team level, promoting collaboration, shared planning, and contextual adaptation. Alternatively, targeted workshops for individual teachers—particularly early-career educators—could provide focused support for adaptive teaching. In both cases, repeated DI-Quest administration could serve as a developmental diagnostic, revealing not only shifts in teacher beliefs but also the evolution of instructional strategies over time. Finally, as DI expands globally, it is essential to explore teachers' experiences within the broader context of educational, historical, political, cultural, and technological factors across different countries. By offering a context-sensitive, mixed-methods analysis of DI

implementation in Türkiye—a country often underrepresented in the international literature—this study contributes valuable insights into how systemic conditions shape teachers' capacity to DI. These findings not only enrich the global conversation on DI but also provide evidence-based guidance for improving policy and practice in comparable educational settings.

CRedit authorship contribution statement

Nilay T. Bümen: Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Hatice Şener Daşcıoğlu:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of Competing Interest

The authors declare no conflicts of interest.

Appendix A. Supporting information

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